ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. There are 594 documents in sci.med.
2. From a data mining perspective isolating specific categories before applying algorithms like frequent pattern mining can be useful because it makes it easier to uncover patterns of each category without mixing the information between categories. It might also be useful if each category has a high volume of information, because then, by first filtering by groups, each algorithm now has an easier workload than before because they do not have to process all data at once.

AI used for this answer: Yes

If yes, how: I read google ai summary to have a general overview of how the frequent pattern mining algorithms work.

External references (links):

- https://www.geeksforgeeks.org/dsa/frequent-pattern-mining-in-data-mining/

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

In this case, if 40% of documents were missing the category_name, dropping these rows will be a bad idea because not only the column is not an independent column, but also because it can be derived by only using the information in the row category. The alternative I would use is to recreate that data by creating by myself a dictionary where I can map the column category and category name. Then, I would create a function that iterates through all the null values in category_name and use the dictionary with the information of the category to give them the name.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The exact category proportions may not always match perfectly the original because as it selects random number it might not select the exact same proportion as the initial dataset. Whereas, as it is not weighted, all the columns have the exact same probability of being selected, then the final proportion will be similar (but probably not the same) as the original.
2. The specific risk of simple random sampling on a highly imbalanced dataset is that if there are low amounts of data of any category it can disappear. In order to control this problem, it is possible to select bigger samples or give more value to this category with a weight.

AI used for this answer: Yes

If yes, how: I used google AI to have an overview of sampling method in Pandas, then I enter to official documentation to understand the parameters.

External references (links):

1. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.sample.html
2. https://www.geeksforgeeks.org/python/python-pandas-dataframe-sample/

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?

2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features increase from 35,481 to 35,788. This makes sense because having the stopwords filter the words selected by the vectorizer.
2. As stopwords are highly common, then adding them will make the percentage of zeros increase, making the matrix less sparse and denser. This is something we want to avoid because these words don't help to differentiate between contexts and having them will increase the data that is being processed while running the algorithms, making an inefficient use of memory and time

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AI used for this answer: Yes

If yes, how: Google AI search for a quick overview of what is a stopword

External references (links):

3. https://docs.nvidia.com/nemo/curator/main/curate-text/process-data/language-management/stopwords

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The total size of the vocabulary is decreased from 0 removed to 33991words removed. This is because in threshold of 0.00 there is nothing to remove and when the threshold is increasing more and more words are being removed.
2. The first words to be removed are words that are only used once. These can be from typos, but also specific terms of words used only by a little amount

of people. These words are good to be eliminated because they don't give much information about the context but do occupy computational resources.

3. Eliminating near-zero variance is a safe way to reduce dimensionality without harming model accuracy because of the properties of the variance. With this I mean when the variance is cero it means that it has no effect on the predictor. Therefore, as near-zero variables may have no effect on the predictor, is a good way to reduce dimensionality. Whereas, it is important to be aware of the context, because sometimes words may have different scales of importance so maybe even though they have low variance they might have a big impact.

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: I tried to ask google search AI to help me to understand why the low variance was safe to remove, but I didn't get the idea until I look in the stack exchange forum.

External references (links):

1. https://stats.stackexchange.com/questions/488989/in-feature-selection-what-is-the-reason-for-considering-removing-low-variance-f

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. For the features sorted with pearson_r the top 10 features are mainly generic words, such as "Hi" or "Thanks". Pearson_r does include some domain-specefic terminology such as "graphics", "video", and "card", but there are not as high ranked as in spearman_r. In spearman_r there are more visible the domain specific words because, having words that did not appear in pearson_r such as "files",

"image", and "3d", although there are some generic words such as "thanks". To sum up, spearman_r not only has more specific terms, but also they are in a higher position.

2. The first difference is that Pearson correlation measures linear correlations using continuous intervals or ration data. Spearson do not need this correlation to be linear, because it measures monotonic relationships in the direction where the variables move. The second is that spearson, as it uses ranks, it can use a different data type such as ordinal, ranked or even interval and ratio data.
3. A specific scenario where Spearman rank correlation with a topic, but a significantly lower Pearson correlation is when the relationship between two variables is monotonic but not linear. An example can be x^3 = y, where Spearman can understand this cubic relationship, but Pearson will not. Another scenario can be when there are outliers, as Spearman rank them, they are not as affected as Pearson, who will try to develop that linear relationship.

AI used for this answer: No

If yes, how:

External references (links):

1. https://www.geeksforgeeks.org/maths/pearson-vs-spearman-correlation-coefficient/

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Decreasing MinSup from 9 to 3 increases the execution time and the number of discovered patterns. For example, in the category comp_graphics it went from a time of 0.08s and 517 patterns to a time of 7.79s and 4854 patterns. This makes sense because at decreasing the threshold there are more elements that must been considered (so the patterns increase) and there is a bigger searching space for the maximal frequent pattern (so the time also increase).

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter F differ on the way they found the patterns, because they try to optimize different things. Top-K FAE search mainly for the most frequent elements of the itemset, which can be either simple words, or combination of words. It doesn't care on the relations between the most frequent, it just care about the frecuency.

On the other hand, minSup in MaxFPGrowth tries to find the maximal frequent patterns that are higher than the MinSup threshold. This means it tries to find the biggest common set that is also frequent, having more context of the words and avoiding having repetitive information. However, it does not have a limit for the patterns found, so it can show lot of unnecessary patterns if the threshold is low.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are structurally more concise than standard frequent patterns because it reduces the number of patterns making the output easier to analyze. It achieves this because it takes the biggest superset that also contains the subsets, so in a way it factorizes the output. And even though it loses the original supports of the subsets, it still includes all the important elements of the itemset.

AI used for this answer: Yes

If yes, how: I got really lost on the notebook explanation so I ask AI to help me to divide each main topic for me to go and search about it.

External references (links):

1. https://www.geeksforgeeks.org/machine-learning/frequent-pattern-growth-algorithm/
2. https://www.reddit.com/r/AIDungeon/comments/1eppgyq/can_someone_explain_what_top_k_and_top_p_are_and/

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. If we want to visually map how these documents group by category on a scatter plot, it is mathematically and visually impossible to plot the raw X_counts matrix directly because it has more than 3 dimensions and it is hard to plot and understand more than 3 dimensions on the same plot.

2. We must do dimensionality reduction to have variables in a space that we can visually understand and without losing information of how data is distributed. Therefore, it is important to use tools like PCA or t-SNE.

AI used for this answer: No

If yes, how:

External references (links):

1. https://www.geeksforgeeks.org/machine-learning/dimensionality-reduction/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not t o over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. The t-SNE shows more distinct and separated cluster islands compared to PCA and UMAP. UMAP still is visible with the differences between the groups but sometimes it feels like it overlaps in the middle. PCA has the worst visual representation because it has all the
2. PCA struggles to separate high-dimensional text categories because it tries to maximize variance of the distribution, but it does not ensure it preserves the local structure. This is different on t-SNE and UMAP, that even though they are more computationally complex, they use some strategies, like using t-distribution in distance, for preserving the dataset local distances.
3. It is important not to over-interpret the physical distance between clusters in a t-SNE plot because the distances do not necessarily represent the real distance because it uses a t-distribution to represent the distance. Therefore, even though visualization might appear to be far away from each point, maybe there are not that separated, and the real distance is not that high.

AI used for this answer: Yes

If yes, how: I used AI to have a general overview on how dimensionality reduction algorithms works.

External references (links):

1. https://www.geeksforgeeks.org/machine-learning/dimensionality-reduction/

2. https://www.geeksforgeeks.org/machine-learning/umap-uniform-manifold-approximation-and-projection/
3. https://www.geeksforgeeks.org/machine-learning/ml-t-distributed-stochastic-neighbor-embedding-t-sne-algorithm/
4. https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. For representing the 4 categories it create 4 columns in a vector where 1 represents it is from that class and 0 represents it is not. For example, for comp.graphics the vector will be [0,1,0,0] because the index that is in 1 represents the class.
2. If an algorithm (like linear regression or KNN) is given the raw integer column (0, 1, 2, 3) as input, it would think a fake order that does not exist because it would think that class 3 is after class 2 and so on.
3. One-hot binarization eliminates this false assumption by generating new columns where it is easier to machines to understand the association because each column represents existence or non-existence of a class, therefore they will not interpret a fake order of the classes.

AI used for this answer: No

If yes, how:

External references (links):

1. https://www.geeksforgeeks.org/machine-learning/ml-one-hot-encoding/
2. https://www.geeksforgeeks.org/machine-learning/k-nearest-neighbours/
